Meridian Mart

Tableau Calculated Fields and Analytical Logic

Purpose: Document the measures and calculation patterns supporting the Meridian Mart dashboards.

1. Commercial Performance Measures

Total Sales

```
SUM([Sales])
```

Adds sales within the current view or filter context. Displayed final KPI: $4.47M.

Total Profit

```
SUM([Profit])
```

Adds the profit values within the current view or filter context. Displayed final KPI: $480.05K.

Profit Margin

```
SUM([Profit])/SUM([Sales])
```

Divides aggregate profit by aggregate sales. Formated as a percentage. Displayed final KPI: 10.74%.

Order Count

```
COUNTD([Order ID])
```

Counts unique order IDs rather than individual item or sales records. Final project value: 2,100 orders.

Usage Percentage

```
COUNTD([Order ID])/TOTAL(COUNTD([Order ID]))
```

A table calculation dividing distinct orders by the total across the visual's partition.

2. Delivery and Shipping Calculations

Delivery Days

```
DATEDIFF("day", [Order Date], [Ship Date])
```

Returns the difference in day boundaries between order and ship dates.

Delvery Time

```
AVG([Ship Date] - [Order Date])
```

An average of underlying date subtraction.

Top Profit by Order

```
{ FIXED [Ship Mode] :
SUM([Profit]) / COUNTD([Order ID])
}
```

Computes profit per distinct order for each shipping mode.

Top profit per Order

```
MAX([Top Profit by Order])
```

Returns the maximum of an internally referenced calculated field (`Top Profit by Order`).

Same Day Delivery

```
{ FIXED [Region] :
    AVG(IF [Ship Mode] = "Same Day"
        THEN [Delivery Days]
    END)
}
```

Calculates an average for Same Day within each region.

Fastest Delivery Range

```
STR(ROUND(MIN([Same Day Delivery]),1))

+

"-"

+

STR(ROUND(MAX([Same Day Delivery]),1))

+

" days"
```

Builds a minimum-to-maximum formatted string in the active visual scope.

3. Subcategory Loss Calculations

Sub-category Profit

```
{ FIXED [Sub-Category] : SUM([Profit]) }
```

Evaluates aggregate profit once per subcategory regardless of visual mark granularity, subject to Tableau's order of operations.

Loss-Making Sub-categories

```
COUNTD(

IF [Sub-category Profit]<0
```

```
THEN [Sub-Category]
END
)
```

Counts distinct subcategories with negative values in the internally referenced profitability calculation.

Combined loss making by profit

```
SUM(
    IF [Sub-category Profit] < 0
    THEN [Profit]
    END
)
```

Sums row-level profit for records belonging to negative-profit subcategories as identified by the internal calculation. The final dashboard shows a combined negative result of approximately $125.24K (reported as a loss magnitude).

4. Discount Analysis Calculations

Average Profit by Discount

```
{ FIXED [Discount (bin)] : AVG([Profit]) }
```

Calculates average profit per underlying record at each discount bin.

Safe Observed Discount

```
MAX(
 IF [Average Profit by Discount]>=0
    THEN [Discount]
    END
)
```

Finds the maximum observed discount among qualifying non-negative average-profit cases.

Discount highlight

```
IF AVG([Discount]) >= 0.24 THEN "High Discount"
ELSE "Normal"
END
```

Flags groups with average discount of at least 24%.

Highest Discount Display

```
IF AVG([Discount]) = WINDOW_MAX(AVG([Discount]))
THEN "Highest"
ELSE "Other"
END
```

Highlights the maximum average discount within the table-calculation partition.

5. View Organisation

Keep Top and Bottom Cities

```
IF [Top 5 Cities] THEN "Top Cities"
ELSEIF [Top 5 bottom Cities] THEN "Bottom cities"
ELSE "Other Cities" END
```

Categorises cities using the workbook’s existing top and bottom flags.